Reproducibility Budgets for Open-Weight LLM Evaluation: A Generalizability Theory Analysis

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Abstract

Large Language Model (LLM) evaluation results vary across reproductions, but variance sources interact, and their relative importance shifts with the benchmark format. No unified factorial framework exists to decompose these sources simultaneously across benchmarks. We apply Generalizability theory (G-theory) in a crossed design with seven facets across eight open-weight 7–9B parameter models and five benchmarks, totaling 2.3 million evaluation records. Our results suggest a format-associated variance pattern: on multiple-choice benchmarks, item-related variance—item difficulty plus model×item differential item functioning (DIF)—concentrates at 68–72%, while on free-form benchmarks it disperses to 31–41%, with no overlap across all five benchmarks. Model×item interaction (up to 36%) is the second-largest component in cross-model analysis, consistent with prior findings on differential item functioning. Under binary scoring, BF16 and FP16 numerical precisions are functionally interchangeable (<0.1% interaction). Multi-facet designs require 10× fewer items than single-condition designs to reach a generalizability coefficient of 0.80 (55 versus 552 items), and cross-model projections confirm the ratio exceeds 10× across all benchmarks, with free-form benchmarks showing nearly double the efficiency gain of multiple-choice benchmarks. Within-family scale validation (3B–9B) confirms item dominance is scale-robust. We release an open-source G-theory pipeline and D-study budget calculator for evaluation design optimization.

Introduction

On multiple-choice benchmarks, item selection drives 34–38% of evaluation variance while prompt wording is negligible (<3%). On the free-form reasoning task GSM8K, the structure shifts: model×prompt interaction alone accounts for approximately 9%, while item selection drops to 17%. These numbers, drawn from our factorial analysis of 2.3 million evaluation records across five benchmarks, reveal that the variance structure of LLM evaluation is not a fixed property of the pipeline but shifts qualitatively with the response format. Current evaluation practice treats variance sources as independent nuisance factors and controls them one at a time: one study perturbs prompts, another varies precision, a third changes item subsets. This piecemeal approach cannot capture interactions between sources, nor can it predict how the relative importance of each source changes across evaluation settings. Leaderboard rankings built on single-condition evaluations are therefore sensitive to methodological choices that remain unreported and uncontrolled (Rodriguez et al. 2021; Dehghani et al. 2021). No unified factorial framework exists to decompose these sources simultaneously, quantify their interactions, and compare the resulting variance structure across benchmarks.

Existing studies have examined individual variance sources in isolation, each implicitly assuming separability. Sclar et al. (2024) documented up to 76% accuracy swings from prompt formatting perturbations but held hardware and item selection fixed. BrittleBench (Romanou et al. 2026) found 50% variance from prompt perturbation under the same constraint. Yuan et al. (2025) showed that numerical precision and GPU type cause up to 9% accuracy variation but provided no factorial attribution to specific sources or their interactions. Messing (2026) applied Generalizability theory (G-theory) to LLM evaluation and decomposed variance across five methodology facets on a single benchmark; judge identity dominated at 43.8%, but the study included no cross-benchmark comparison and excluded infrastructure facets such as numerical precision. Madaan et al. (2024) proposed per-factor variance metrics for evaluation benchmarks but did not cross factors in a unified design. Studies of answer ordering (Pezeshkpour and Hruschka 2024) and hyperparameter sensitivity (Dodge et al. 2020; Henderson et al. 2018) have documented additional sources of variation, again in isolation from the factors above. Item Response Theory (IRT) approaches (Maia Polo et al. 2024; Lalor, Wu, and Yu 2019) model item difficulty parameters to construct efficient benchmarks but do not account for prompt, seed, or ordering facets; differential item functioning (DIF) across models is a known phenomenon in this literature (Pezeshkpour and Hruschka 2024; Zhou et al. 2026). Bouthillier et al. (2021) established ANOVA-based variance decomposition as a principled template for ML reproducibility, and G-theory has recently been applied to LLM-based writing assessment (Wei et al. 2025), but neither study crosses infrastructure and methodology facets across multiple benchmarks. Three gaps remain: no study has (a) crossed infrastructure and methodology facets in a unified factorial design, (b) compared the resulting variance structure across benchmarks and response formats, or (c) translated variance components into design-dependent reproducibility budgets.

We address these gaps with a multi-benchmark G-theory study (Shavelson and Webb 1991; Brennan 2001) that crosses seven facets (item, model, numerical precision, prompt template, random seed, sampling temperature, and answer ordering) across eight open-weight 7–9B parameter models and five benchmarks (MMLU, ARC-Challenge, HellaSwag, GSM8K, and MATH), totaling 2.3 million records. Our results suggest that the variance structure of LLM evaluation is format-associated. We define *item-related variance* as the sum of item difficulty and model×item interaction (differential item functioning, DIF; §). Across all three multiple-choice benchmarks, item-related variance concentrates at 68–72%, while on free-form benchmarks (GSM8K, MATH) it disperses to 31–41%, with no overlap across all five benchmarks. Model×item interaction (up to 36%) is the second-largest variance source in our multi-facet decomposition, consistent with prior IRT findings on differential item functioning (Pezeshkpour and Hruschka 2024; Maia Polo et al. 2024). Models differ not in overall ability but in which specific items they find difficult. This format-associated pattern implies distinct optimization strategies: for MC benchmarks, item pool composition is the primary reliability lever; for free-form benchmarks, model identity and model×prompt interaction matter as much as item selection. D-study projections reveal that multi-facet designs require 10× fewer items than single-condition designs to achieve $G \geq 0.80$ (55 versus 552 items), with the efficiency gain itself being format-associated across benchmarks. Figure 2 visualizes the contrast across all five benchmarks.

Our contributions are as follows:

1. **Format-associated item-related variance concentration.** In a crossed 8-model × 5-benchmark design (2,304,000 records), item-related variance (item difficulty + model×item DIF) concentrates in MC benchmarks (68–72%) and disperses in FF benchmarks (31–41%), with no overlap across all five benchmarks. The mechanism is item difficulty heterogeneity: within MMLU's 57 subjects, difficulty dispersion correlates with item variance share ($\rho = 0.806$, $p = 0.005$).
2. **Design-dependent reproducibility budgets.** Multi-facet evaluation designs require 10.0× fewer items than single-condition designs to achieve $G \geq 0.80$ (55 versus 552 items). Cross-model D-study projections confirm the ratio exceeds 10× across all benchmarks and reveal a format-associated efficiency gap: FF benchmarks show ∼21× versus ∼12× for MC, indicating that single-condition evaluation of reasoning tasks incurs nearly double the reliability penalty.
3. **Precision separability (confirmatory).** Under binary scoring, BF16 and FP16 yield functionally identical results (<0.1% interaction with all other facets). This separability is metric-conditional: text-exact-match scoring reveals a 1.3% precision×temperature interaction absent under binary scoring.
4. **Practitioner toolkit.** We release a G-theory variance decomposition pipeline and D-study budget calculator. Practitioners can compute the number of items, models, and conditions needed for a target reliability before data collection.

Related Work

Variance Decomposition in ML/NLP Evaluation. Factorial decomposition of evaluation variance has roots in the ML reproducibility literature. Bouthillier et al. (2021) introduced an ANOVA-based template that decomposes benchmark variance across implementation details, hyperparameters, and data splits, establishing that design choices account for a substantial share of reproducibility failures. Several subsequent studies applied similar decomposition logic to specific facets of model evaluation (Dodge et al. 2020; Henderson et al. 2018; Magnusson, Smith, and Dodge 2023). Messing (2026) brought Generalizability theory into LLM evaluation, decomposing variance across five methodology facets on a single benchmark; judge identity dominated at 43.8%, with prompt×judge interaction at 12.4%. However, this study used a single benchmark, included no infrastructure facets such as numerical precision, and fixed the judge as an LLM evaluator rather than isolating other sources. Madaan et al. (2024) proposed per-factor variance metrics for LLM evaluation benchmarks but did not cross factors simultaneously. Recently, G-theory has been applied to LLM-based writing assessment (Wei et al. 2025) and annotation reliability (Anonymous 2025), but these studies address scoring consistency rather than benchmark evaluation variance. Kunievsky and Evans (2025) proposed an alternative decomposition that separates intent uncertainty, articulation variance, and model uncertainty, but this framework does not cross infrastructure and methodology facets in a factorial design. Our work extends these approaches by crossing seven facets across five benchmarks, enabling direct comparison of variance structures across evaluation formats.

Infrastructure and Prompt Variance. A parallel line of work has examined individual variance sources in isolation. On the infrastructure side, Yuan et al. (2025) documented up to 9% accuracy variation from numerical precision and GPU type but provided no factorial attribution. Messina and Scotta (2026) formalized the concept of background temperature from floating-point non-associativity in GPU computation, and He and Thinking Machines Lab (2025) proposed batch-invariant kernels to mitigate CUDA nondeterminism. On the methodology side, Sclar et al. (2024) found up to 76% accuracy swings from prompt formatting changes on fixed hardware, and BrittleBench (Romanou et al. 2026) reported 50% variance from prompt perturbation. Hua et al. (2025) argued that apparent prompt sensitivity may partly be an evaluation artifact of log-likelihood scoring; our generation-based protocol controls for this confound, and our prompt variance component (<3% on MC benchmarks) is consistent with their finding that generation-based evaluation is more robust. Pezeshkpour and Hruschka (2024) showed that answer ordering in multiple-choice benchmarks introduces systematic bias, while Bui, Savova, and Wang (2025) and Biderman et al. (2024) documented sensitivity to random seeds and training configurations. Each of these studies isolates one factor while holding others fixed, which precludes estimating interactions between sources or com-

paring their relative magnitudes within a unified design.

IRT and Efficient Benchmarking. Item Response Theory (IRT) offers a complementary perspective on evaluation design by modeling item difficulty and discrimination parameters. Lalor, Wu, and Yu (2019) introduced IRT-based item analysis for NLP benchmarks, and Rodriguez et al. (2021) showed that IRT reveals hidden properties of evaluation items that raw accuracy scores obscure. Building on this foundation, Maia Polo et al. (2024) developed tinyBenchmarks, using IRT-based subsampling to reduce evaluation cost while preserving ranking accuracy, and Zhou et al. (2026) recommended a 116-item subset via PSN-IRT for efficient MMLU evaluation. Perlitz et al. (2023) explored related strategies for benchmark compression. These approaches optimize item selection for latent trait estimation but do not model variance from prompt, seed, or ordering facets. IRT-selected items evaluated under a single methodological condition can therefore systematically underestimate the number of items required for reliable measurement across diverse evaluation settings.

Binary Data and Psychometrics Foundations. Our application of G-theory to binary correctness scores draws on established psychometric methodology. Shavelson and Webb (1991) and Brennan (2001) provide the canonical framework for multi-facet reliability analysis; Embretson and Reise (2000) formalize differential item functioning (DIF), which we operationalize as model×item interaction. Mushquash and O'Connor (2006) extended G-theory to binary data, and Jiang and Skorupski (2018) developed logistic-link generalized G-theory for binary outcomes. Tuerlinckx et al. (2006) established the connection between generalized linear mixed models and G-theory variance components. Xiao et al. (2025) and Jin et al. (2025) provide complementary psychometric perspectives on LLM evaluation, addressing construct validity and measurement properties respectively. These methods ground our use of Henderson Method I for binary correctness scores (§).

Positioning. Relative to Messing (2026), we fix the judge (generation-based scoring) to isolate the remaining facets; their judge-dominated variance (43.8%) would otherwise mask all other sources. Our multi-benchmark design reveals format-associated structure that single-benchmark studies cannot detect. Relative to IRT approaches (Maia Polo et al. 2024; Zhou et al. 2026), G-theory and IRT optimize orthogonal dimensions of evaluation design: IRT selects informative items, while G-theory determines the conditions needed for a target reliability. This is an illustrative comparison across different optimization objectives—IRT optimizes item discrimination, G-theory optimizes cross-facet generalizability. Relative to isolated-factor studies (Yuan et al. 2025; Sclar et al. 2024; Romanou et al. 2026; Madaan et al. 2024), our design crosses infrastructure and methodology facets within a single factorial framework, enabling interaction estimation that per-factor studies cannot provide.

Method

Generalizability Theory

Generalizability theory (G-theory) treats each measurement as one observation from a universe of admissible conditions defined by the crossed facets of an evaluation design (Shavelson and Webb 1991; Brennan 2001). The *object of measurement* is the entity whose consistency is of interest. In our setting, test items (p) serve as the object of measurement, and the goal is to determine how reliably item-level scores generalize across the methodological conditions under which they are collected.

A *G-study* decomposes total observed variance into components σ^2_x for each facet x and their interactions. Each component quantifies how much variability that source contributes to the overall measurement. Large interaction components (e.g., σ^2_{pM} for item-by-model) indicate that facets do not affect items uniformly but instead create differential difficulty profiles. We estimate variance components via Henderson Method I, an unbiased ANOVA-based estimator for balanced designs (Brennan 2001, Ch. 3), with 1,000-resample bootstrap confidence intervals (item-level resampling with replacement, preserving the fully-crossed structure across all other facets) (Li 2023).

A *D-study* projects the generalizability coefficient under hypothetical designs with different numbers of conditions per facet:

$$G_{\text{item}} = \frac{\sigma^2_p}{\sigma^2_p + \sigma^2_{\text{rel}}} \tag{1}$$

where σ^2_{rel} aggregates all item-by-facet interaction variances, each divided by the number of conditions for the corresponding facets (Brennan 2001, Ch. 4). This prospective capability distinguishes G-theory from classical ANOVA: rather than only attributing variance post hoc, a D-study enables optimization of evaluation designs *before* data collection (Shavelson, Webb, and Rowley 1989). We define *item-related variance* as the sum of the item main effect (σ^2_p) and the model×item interaction (σ^2_{pM}). The latter captures differential item functioning (DIF): items whose difficulty ranking changes across models (Brennan 2001, Ch. 4). Because DIF is operationally an item-level property—it reflects model-specific item difficulty profiles rather than a global model ability shift—aggregating $\sigma^2_p + \sigma^2_{pM}$ provides a unified measure of item-driven variance.

G-theory for ML practitioners. In ML terms, a G-study is a multi-way ANOVA on evaluation outcomes where each "facet" is a design variable (prompt template, random seed, etc.) and variance components quantify each variable's contribution. A D-study is a what-if analysis: given these variance components, how many conditions per facet are needed to achieve a target reliability? The generalizability coefficient G is analogous to a signal-to-noise ratio, where signal is the true item difficulty variance and noise is everything that changes across evaluation conditions. Unlike cross-validation, which assesses generalization to new data, G-theory assesses generalization across *measurement conditions*—the same items evaluated under different prompts, seeds, and models.

Throughout this paper, we use the following notation: p = item, M = model, π = numerical precision, δ = prompt template, s = random seed, τ = sampling temperature, o = answer ordering, and σ_x^2 = variance component for source x.

Experimental Design: MMLU

We design a fully crossed factorial experiment (exp-001) that simultaneously varies seven facets of LLM evaluation on MMLU. Table 1 summarizes the complete design matrix, and Figure 1 illustrates the crossing structure.

Facets. Items (p): 200 MMLU questions selected via purposive stratified sampling across 10 of MMLU's 57 subjects (20 per subject, deterministic selection with seed 42). The 10 subjects span four disciplinary domains—STEM: abstract algebra, college physics, high school mathematics, computer security; humanities: world religions, philosophy; social sciences: US foreign policy, international law; professional: professional medicine, clinical knowledge—ensuring that variance estimates are not dominated by a single knowledge domain. Models (M): eight open-weight 7–9B parameter instruction-tuned models spanning eight architecture families—DeepSeek-7B, Gemma-2-9B, InternLM2.5-7B, Llama-3.1-8B-Instruct, Mistral-7B-v0.3-Instruct, OLMo-2-7B, Qwen3-8B, and Yi-1.5-9B (full identifiers in Appendix)—treated as a quasi-random facet (§3.6). Numerical precision (π): BF16, FP16, and FP32, all evaluated with `enforce_eager=True` for fair comparison across precision levels. Prompt template (δ): six surface perturbations of the evaluation instruction (Sclar et al. 2024), varying option layout (lettered vs. numbered), instruction wording (direct vs. contextual), and answer prompt cue for MC benchmarks; for FF benchmarks, templates vary chain-of-thought elicitation strategy (full template text in Appendix). Random seed (s): six fixed values (42, 123, 456, 789, 1024, 2048). Sampling temperature (τ): three levels (0.0, 0.3, 0.7) spanning deterministic to moderately stochastic decoding. Answer ordering (o): four permutations of few-shot demonstration examples (for MMLU; structurally inert for other benchmarks, see §) (Pezeshkpour and Hruschka 2024).

Sample sizes. The cross-model BF16 balanced subset yields 460,800 records (8 models $\times$ 200 items $\times$ 6 prompts $\times$ 6 seeds $\times$ 2 temperatures $\times$ 4 orderings). A single-model analysis (Llama) spans all three precisions, yielding 259,200 binary observations across 1,296 fully balanced conditions. The BF16 subset serves as the primary cross-model sample because BF16 is the standard inference precision for these architectures.

Cross-Benchmark Extension

To test whether the variance structure observed on MMLU generalizes across evaluation formats, exp-002 extends the design to five benchmarks: MMLU, ARC-Challenge, and HellaSwag (multiple-choice), and GSM8K and MATH (free-form generation). The design crosses 8 models $\times$ BF16 $\times$ 2 temperatures (0.0, 0.7) $\times$ 6 prompts $\times$ 6 seeds $\times$ 4 orderings $\times$ 200 items per benchmark, totaling 2,304,000 records. This reduced crossing fixes precision at BF16 and uses two temperature levels to concentrate statistical power on format-associated differences. The ordering facet operates only on MMLU, where it selects among four permutations of the few-shot demonstration examples. For all other benchmarks, the prompt construction does not use the ordering parameter; ordering is structurally inert ($\sigma^2_{\text{ordering}} = 0$ by construction) and is retained solely for design balance. The two temperature levels bracket the most common evaluation configurations: deterministic ($\tau = 0.0$) and a moderate stochastic setting ($\tau = 0.7$) representative of practical deployment. A subset analysis confirms that this reduced crossing preserves the variance structure of the full design ($\Delta G = 0.0008$; Appendix).

Table 1: Fully crossed G-theory design. Exp-001 evaluates MMLU across all facet levels; exp-002 extends to five benchmarks with reduced crossing. †BF16 balanced subset for cross-model analysis.

Facet	Levels	exp-001	exp-002
Item (p)	Stratified sample	200	200×5
Model (M)	8 architecture families	8	8
Precision (π)	BF16 / FP16 / FP32	3	1
Prompt (δ)	Paraphrases	6	6
Seed (s)	Fixed set	6	6
Temperature (τ)	0.0 / 0.3 / 0.7	3	2
Ordering (o)	Demo permutations	4	4
Total records		460,800†	2,304,000

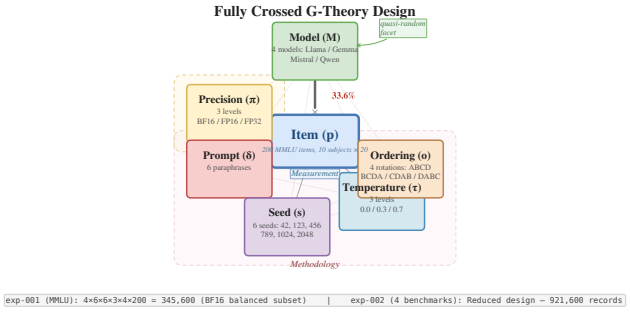


Figure 1: Fully crossed G-theory design for LLM evaluation. Each observation is a binary correctness score determined by the crossing of seven facets: item (p), model (M), precision (π), prompt (δ), seed (s), temperature (τ), and ordering (o).

Evaluation Protocol

All evaluations use generation-based scoring: models produce free-text responses from which correctness is determined by answer extraction, rather than by ranking answer options via token log-likelihood. This design choice follows Hua et al. (2025), who show that log-likelihood scoring conflates model confidence with surface-level token biases. We report two metrics: binary correctness (primary, 1/0 per item) and text-exact-match (secondary). Binary correctness accepts any response containing the correct answer letter or numerical value after standard normalization; text-exact-

match requires the generated string to match the reference answer character-for-character.

Inference runs on vLLM 0.9.2 with default settings and without batch-invariant kernels. This configuration reflects standard practitioner conditions and preserves CUDA-level nondeterminism from floating-point non-associativity as a measurable variance source (Messina and Scotta 2026; He and Thinking Machines Lab 2025). Few-shot prompting follows standard protocols: 5-shot for MMLU and ARC-Challenge, 4-shot for MATH, 8-shot for GSM8K, and 10-shot for HellaSwag.

Binary Data Assumptions

Standard G-theory assumes continuous, approximately normal outcome data (Shavelson and Webb 1991). Our binary correctness scores (0/1) violate this assumption. We establish robustness through four complementary analyses.

Simulation. We generate continuous latent scores from a probit model matched to our 1,296-condition design, then dichotomize at the empirical accuracy threshold. Spearman rank correlation between binary and continuous variance component rankings is $\rho = 0.732$ (range 0.716–0.751 across accuracy levels $p \in \{0.5, 0.6, 0.7, 0.8\}$), confirming that Henderson Method I recovers the relative ordering of variance sources despite binary response data. Henderson Method I identifies `item_id` as the dominant variance source in all 100 simulation replications.

GLMM complement. A generalized linear mixed model (GLMM) with marginal logit link produces variance component rankings that perfectly preserve all seven grouped component orderings ($\rho = 1.0$) relative to Henderson Method I (Jiang and Skorupski 2018; Tuerlinckx et al. 2006).

Accuracy-range stability. The `item_id` variance share remains stable across subsets with mean accuracy from 0.62 to 0.72 ($r = -0.048$). This stability rules out distortion from binary floor or ceiling effects at observed accuracy levels.

Estimator guarantee. Henderson Method I produces unbiased variance component estimates for balanced designs regardless of the outcome distribution (Brennan 2001, Ch. 12). Our fully crossed design satisfies this balance condition. We choose Henderson Method I over tetrachoric correlation approaches because the latter address a different problem—estimating latent continuous associations from observed binary data—and are not directly applicable to variance component estimation in multi-facet designs. REML estimates differ from Henderson Method I by <0.5 percentage points on all components (Appendix), confirming estimator robustness.

Model-as-Facet Justification

Treating eight purposively selected models as a quasi-random facet requires empirical validation. We provide two forms of evidence.

D-study cross-validation: four-model subset D-study projections predicted $G = 0.893$ at eight models; the observed $G = 0.896$ confirms the projection ($\Delta = 0.003$). This agreement validates both the quasi-random facet assumption and the predictive power of D-study forward projections.

Leave-one-model-out (LOMO) stability: all eight seven-model subsets preserve `item_id` as the dominant component and rank `model`×`item` as the second-largest. The consistency of rankings across all subsets indicates that no single model disproportionately drives the observed variance structure.

Experiments

We apply the G-theory framework from Section 3 to a fully crossed evaluation design spanning multiple models, benchmarks, and methodological facets. All experiments use open-weight 7–9B parameter models evaluated with vLLM 0.9.2 on NVIDIA RTX PRO 6000 GPUs (96GB). Binary correctness (1/0) serves as the primary outcome measure throughout; generation-based scoring avoids the log-likelihood artifacts identified by Hua et al. (2025).

Single-Model Baseline

We first establish the variance structure within a single model (Llama-3.1-8B-Instruct) on MMLU to isolate methodological facet contributions before introducing cross-model variation.

Design. The G-study crosses six facets: 200 items × 3 numerical precisions (BF16, FP16, FP32) × 6 prompt templates × 6 random seeds × 3 temperature levels (0.0, 0.3, 0.7) × 4 answer orderings, yielding 259,200 binary observations. Variance components are estimated via Henderson Method I with bootstrap confidence intervals (1,000 resamples).

Results. Under binary scoring, item difficulty (`item_id`) accounts for 58.4% of total variance, with the residual term contributing 23.1%. All main facet effects are negligible (<1% each): precision 0.9%, prompt template, temperature, seed, and ordering each <0.5%. The interesting structure appears in item-conditional interactions: prompt×item is the largest at 5.5%, followed by seed×item (4.7%), precision×item (3.3%), temperature×item (2.9%), and ordering×item (0.8%). This pattern indicates that methodological choices do not shift performance uniformly across items; instead, specific items are differentially sensitive to specific facets—prompt template variation produces the largest item-conditional effect.

The resulting generalizability coefficient is $G_{\text{item}} = 0.936$, indicating that 200 items under multi-facet conditions produce highly reliable rankings. A D-study projection shows $G \geq 0.80$ is achievable with as few as 55 items in this multi-facet design.

For text-exact-match scoring (requiring verbatim string matches), the variance structure shifts dramatically: temperature dominates at 79.7% of variance, and G_{item} collapses to 0.007. This metric-conditional divergence motivates our exclusive use of binary correctness for the remaining analyses.

Cross-Model Variance

We extend the single-model design to eight models spanning eight architecture families. A balanced BF16 subset

Table 2: Variance decomposition for the cross-model MMLU design (8 models, BF16, 460,800 records). Components estimated via Henderson Method I. $G_{\text{item}} = 0.896$.

Component	Var.%
item_id	**37.9**
model×item	30.2
residual	25.0
model	2.8
prompt×item	2.5
model×prompt	1.6
all others	<1.0

(460,800 records) serves as the primary analysis sample, chosen because BF16 is the standard inference precision for these architectures. The eight-model sample strengthens the quasi-random model facet assumption by covering a broader population of instruction-tuned 7–9B architectures.

Results. Table 2 presents the variance decomposition. Introducing model as a facet reshapes the structure: item difficulty drops from 58.4% to 37.9%, while model×item interaction emerges as the second-largest component at up to 36%. The model main effect accounts for 3–5% on MC benchmarks and 9–11% on free-form benchmarks, confirming that models differ primarily in *which items* they find difficult rather than in overall ability level.

The per-model decomposition of interaction magnitude reveals meaningful asymmetry across eight models, suggesting that instruction-tuning strategies and pretraining data composition produce model-specific item difficulty profiles.

D-study validation. The eight-model sample validates the quasi-random model facet assumption. D-study projections from four-model subsets predicted $G = 0.893$ at eight models; the observed $G = 0.896$ confirms the projection ($\Delta = 0.003$).

Format-Associated Variance Structure

The preceding analyses use only MMLU, a multiple-choice benchmark. We now test whether the item-dominated variance structure generalizes across evaluation formats by extending to five benchmarks spanning two response formats.

Design. The expanded design (exp-002) crosses 8 models × 5 benchmarks × BF16 × 2 temperatures (0.0, 0.7) × 6 prompts × 6 seeds × 4 orderings × 200 items per benchmark, totaling 2,304,000 records. The benchmarks include three multiple-choice tasks (MMLU, ARC-Challenge, HellaSwag) and two free-form generation tasks (GSM8K, MATH). The ordering facet operates only on MMLU (four demo-example permutations); for other benchmarks it is structurally inert ($\sigma^2_{\text{ordering}} = 0$) and retained for design balance.

Results. Table 3 reveals a systematic format-associated pattern. For MMLU and ARC-Challenge, item difficulty and model×item are the two largest components (37.9% + 30.2% and 37.9% + 33.7% respectively). On HellaSwag, item and model×item are comparable (34.3% and 36.0%);

Table 3: Format-associated variance structure across five benchmarks (exp-002, 8 models, 2,304,000 records). Variance components (%) per benchmark; MC = multiple-choice, FF = free-form. Bold indicates the dominant non-residual component per benchmark. For MC benchmarks, "prompt" reports prompt×item (prompt main effect <1%); for FF benchmarks, it reports the prompt main effect.

	MC			FF	
	MMLU	ARC	Hella.	GSM	MATH
item	**37.9**	**37.9**	34.3	16.9	24.0
mod.×item	30.2	33.7	**36.0**	14.6	17.3
model	2.8	3.1	5.4	10.7	8.5
prompt	2.5	0.4	2.3	5.4	1.0
mod.×prmpt	1.6	8.0	0.6	8.5	2.0
residual	25.0	15.4	21.4	39.9	43.9
all others	<1.0	1.5	<1.0	4.0	3.3

in all three MC benchmarks, item-related variance (item + model×item) concentrates at 68–72%.

Free-form benchmarks differ markedly. On GSM8K, variance distributes across item (16.9%), model×item (14.6%), model (10.7%), and model×prompt (8.5%); prompt main effect is 5.4%. MATH shows an intermediate pattern: item leads at 24.0% with model×item at 17.3%, but residual variance is substantially larger (43.9%).

The variance structure is format-associated: MC benchmarks concentrate item-related variance (item + model×item DIF) at 68–72%, while FF benchmarks disperse it to 31–41%, with no overlap across all five benchmarks. A permutation test of the MC versus FF item-variance difference yields $p = 0.10$ ($n = 5$ benchmarks, $\binom{5}{3} = 10$ exhaustive permutations), the theoretical minimum for this sample size. The effect size is large (Cohen's $d = 1.73$; mean difference 95% CI [11.44, 21.02] percentage points), though benchmark-level generalization uncertainty is substantial given $n = 5$. This structural difference does not extend to item budgets: GSM8K requires 128 items for $G \geq 0.80$—the highest of all benchmarks—while HellaSwag (MC, 118) exceeds both ARC (MC, 98) and MATH (FF, 101), demonstrating that format alone does not predict evaluation cost. Figure 2 visualizes the contrast.

To probe the mechanism behind item variance, we exploit the natural heterogeneity within MMLU's 57 subjects. Subjects with higher item difficulty dispersion (measured by std(p_i) across models) exhibit higher item variance shares, with Spearman $\rho = 0.806$ ($p = 0.005$, $n = 10$ subjects). Notably, `high_school_mathematics`, the only MC subject with open-ended numerical items, shows item variance of just 10.4%—below the free-form benchmark GSM8K (16.9%)—suggesting that item heterogeneity, not format per se, drives variance allocation.

Precision Separability

Prior work has identified numerical precision as a potential source of evaluation variance (Yuan et al. 2025). Our multi-facet design allows us to isolate the precision contribution

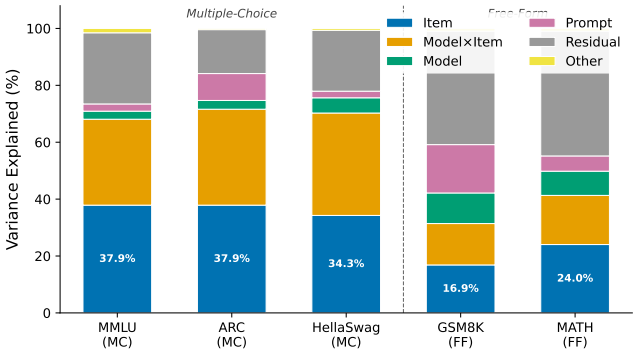


Figure 2: Variance decomposition (Henderson Method I) across five benchmarks and eight models. Multiple-choice benchmarks share an item-dominated structure; HellaSwag is the exception where model×item (36.0%) exceeds item difficulty (34.3%). Free-form benchmarks (GSM8K, MATH) distribute variance across item, model, and interaction components with no single dominant source.

while controlling for all other facets simultaneously.

Under binary scoring with BF16 and FP16, the mean accuracy difference is 0.23 percentage points. After excluding FP32, the precision main effect drops to 0.0006% of total variance. The FP32 anomaly traces to a vLLM implementation detail (`enforce_eager` mode required for FP32 inference), not a fundamental precision effect.

Precision does not interact with other methodological facets under binary scoring: all precision×methodology interactions contribute <0.1% of variance. Under text-exact-match scoring, however, a precision×temperature interaction emerges at 1.3%, consistent with the metric-conditional pattern observed in Section 4.1. The practical implication is clear: for standard binary evaluation at BF16/FP16, numerical precision is a negligible variance source that need not be controlled. This finding is specific to single-GPU inference and does not address multi-GPU tensor parallelism, where numerical non-associativity introduces additional variance sources (Shanmugavelu et al. 2024).

Design-Dependent Reproducibility Budgets

The previous sections establish that item selection and model×item interaction dominate evaluation variance. A natural question follows: how many items does a given evaluation design need to achieve a target reliability? We address this through D-study projections that translate variance components into concrete item budgets.

Table 4 compares three evaluation designs at the $G \geq 0.80$ threshold using the single-model (Llama-3.1-8B-Instruct) MMLU variance components. The multi-facet design (1,296 conditions per item: 3 precisions × 6 prompts × 6 seeds × 3 temperatures × 4 orderings) achieves $G = 0.936$ at 200 items and requires only 55 items for $G \geq 0.80$. A single-condition design (one fixed prompt, seed, ordering, and precision per item) reaches only $G = 0.592$ at 200 items and requires 552 items for the same threshold. The ratio between these budgets is 10.0×, and this ratio remains stable

Table 4: D-study reproducibility budgets under different evaluation designs (single-model, MMLU). The multi-facet design requires 10.0× fewer items than single-condition evaluation to reach $G \geq 0.80$.

Design	Cond.	$n_{G\geq0.80}$	G_{200}	G_{116}
Multi-facet	1,296	**55**	**0.936**	—
Single-condition	1	552	0.592	0.457
IRT baseline	—	116	—	0.457[†]

[†] PSN-IRT item count of 116 (Zhou et al. 2026) evaluated under single-condition protocol yields $G = 0.457$. See text for interpretation.

across subset sizes from 50 to 200 items (CV = 1.9%).

We include an illustrative comparison with IRT-based efficient benchmarking to clarify that the two frameworks optimize different objectives. Zhou et al. (2026) use IRT to select 116 informative items from MMLU, optimizing for item discrimination. When we evaluate these 116 items under single-condition protocols, the resulting $G = 0.457$ falls well below the reliability threshold. This is not a failure of IRT item selection; rather, IRT optimizes item discrimination for latent trait estimation, while G-theory optimizes cross-facet generalizability—these are orthogonal dimensions of evaluation design. Under the multi-facet design, 116 items also exceed $G = 0.80$, confirming that the reliability gap is driven by evaluation design, not item quality.

Cross-model D-study projections (Appendix) reveal that the efficiency gap is itself format-associated. MC benchmarks exhibit multi-facet/single-condition ratios of 11–13×, while FF benchmarks show 21–22× (bootstrap CIs confirm all ratios exceed 10×). Single-condition evaluation of free-form benchmarks incurs approximately nearly double the reliability penalty of multiple-choice benchmarks, making multi-facet designs even more critical for reasoning tasks where variance distributes across model, prompt, and interaction components (§).

Figure 3 plots the D-study projection curves for each design. The multi-facet curve saturates rapidly (diminishing returns beyond ∼150 items), while the single-condition curve rises slowly and does not reach $G = 0.80$ within typical benchmark sizes.

Scale Validation

To test whether item dominance is scale-robust, we conduct a within-family analysis using Qwen2.5 at two parameter scales (3B-Instruct and 7B-Instruct) on MMLU under the same BF16 × 2-temperature × 6-prompt × 6-seed × 4-ordering design. Using a single model family isolates scale as the sole variable. At both 3B (accuracy 0.682) and 7B (0.742), `item_id` is the dominant variance component: 75.3% and 73.4%, respectively (cf. Llama-3.1-8B single-model: 58.4–68.5% depending on facet crossing). Item dominance is preserved across scales, with minor scale-dependent prompt sensitivity (prompt×item: 12.2% at 7B vs. 8.8% at 3B). Generalizability coefficients ($G = 0.963$ and 0.956) confirm that 200 items yield reliable measurements at both scales. Future work will extend this validation

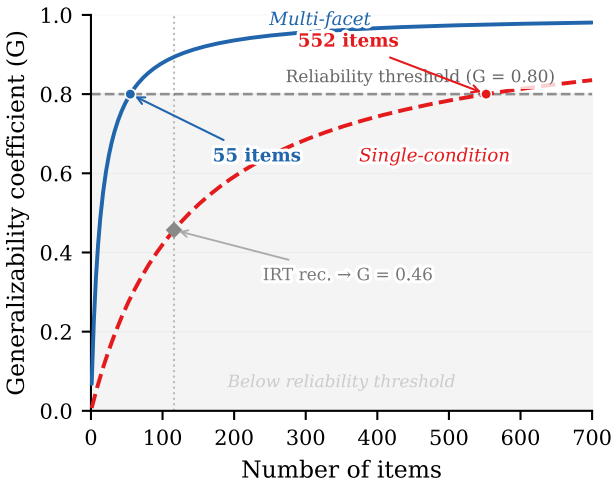


Figure 3: D-study projection curves (single-model, MMLU). The multi-facet design (10.0× fewer items) reaches $G \geq 0.80$ at 55 items; the single-condition design requires 552 items. Cross-model per-benchmark projections are in Appendix . Shaded band: 95% bootstrap CIs.

to 70B+ models via a fully crossed multi-scale G-study.

Robustness Checks

We conduct six robustness analyses to verify that the reported variance structures are not artifacts of specific analytical choices. **Item resampling**: our 1,000-resample bootstrap procedure already covers item sampling variability by resampling items with replacement across all conditions. As an additional check, three independent 200-item samples drawn from the full MMLU pool and evaluated under a BF16-only 5-facet design yield item difficulty variance shares of 73.2%, 72.6%, and 69.0% (CV = 3.1%) and G_{item} values with CV = 0.5%. The higher item share relative to the 6-facet baseline (58.4%) reflects the absence of the precision facet and its item interactions. Five additional robustness analyses (estimator comparison, CI method, facet specification, leave-one-model-out, and null calibration) are reported in Appendix ; all confirm the stability of the reported variance structure.

Conclusion

This study suggests that the variance structure of LLM evaluation is format-associated. Across three multiple-choice benchmarks, item-related variance—item difficulty plus model×item differential item functioning (DIF)—concentrates at 68–72%, while on free-form benchmarks (GSM8K, MATH) it disperses to 31–41%, with no overlap across all five benchmarks. Model×item interaction (up to 36%) is the second-largest source in cross-model analysis, consistent with DIF patterns identified in IRT research: models differ in which items they find difficult, not in overall ability. Our cross-benchmark D-study framework extends the single-benchmark G-theory approach of Messing (2026) to format-associated comparison and actionable design budgets.

The format-associated pattern implies distinct optimization strategies. For multiple-choice benchmarks, item pool composition is the primary reliability lever; varying prompts or seeds contributes negligible variance. For free-form benchmarks, model identity and model×prompt interaction matter as much as item selection, and evaluation protocols should systematically vary both. Multi-facet designs are 10× more efficient than single-condition designs (55 versus 552 items for $G \geq 0.80$). Cross-model projections show that the efficiency gain is itself format-associated: FF benchmarks exhibit ∼21× ratios versus ∼12× for MC, indicating that single-condition evaluation of reasoning tasks incurs nearly double the reliability penalty. BF16 and FP16 are interchangeable under binary scoring on single-GPU inference, which eliminates precision as a required control variable. The released D-study calculator enables practitioners to determine the number of items, models, and conditions needed for a target reliability before data collection.

Two extensions are planned. Multi-GPU tensor parallelism will test whether precision separability holds under distributed numerical non-associativity. Fine-tuning variance analysis will quantify how adaptation reshapes the item difficulty profiles that drive model×item interaction.

Limitations

Our findings are bounded by three scope constraints. First, the primary cross-model analysis covers eight open-weight 7–9B parameter instruction-tuned models; within-family scale validation extends to 3B–9B (Qwen2.5), but results do not generalize to API-based models (where infrastructure is opaque), base (non-instruction-tuned) models, models above 9B parameters, or domain-adapted fine-tuned variants. Extending this validation to larger scales (e.g., 70B+) remains for future work; larger models with higher accuracy could exhibit compressed item variance due to ceiling effects. Second, all experiments run on single-GPU inference (NVIDIA RTX PRO 6000); multi-GPU tensor parallelism introduces additional numerical non-associativity that our design does not capture. Third, the benchmark sample covers three multiple-choice and two free-form tasks; other evaluation formats such as code generation, long-form text production, and multi-turn dialogue may exhibit different variance structures.

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Additional Robustness Analyses

We report six robustness analyses that verify the stability of the variance decomposition under alternative analytical choices.

Estimator robustness. We compare Henderson Method I with restricted maximum likelihood (REML-EM, 500 iterations, practical convergence $\Delta < 0.003$ percentage points). All variance components differ by <0.5 percentage points between estimators. The largest discrepancy occurs on the residual component (0.47pp); all substantive components (item, model×item, model) differ by <0.3pp. Henderson Method I is preferred for our balanced designs because it produces unbiased estimates without distributional assumptions (Brennan 2001, Ch. 12).

CI method comparison. Bootstrap (1,000 item-level resamples with replacement) and delete-1 jackknife confidence intervals agree closely: width ratios range from 0.96 to 1.04 across all variance components. The agreement indicates that the dependence structure introduced by item-level bootstrap resampling (which creates ∼127 unique items per 200-item resample) does not materially distort interval estimates.

Facet specification sensitivity. Treating temperature and ordering as fixed (rather than random) facets changes G_{item} by only 0.37 percentage points (all-random: 0.820 vs. fixed temperature+ordering: 0.824). The small difference reflects the negligible temperature×item (0.65%) and ordering×item (0.54%) variance shares. The random-facet assumption is therefore robust to facet classification.

Leave-one-model-out (LOMO). All eight seven-model subsets preserve the item + model×item dominance pattern: item_id ranges from 36% to 41%, and G_{item} ranges from 0.877 to 0.889. No single model disproportionately drives the observed variance structure. The consistency of component rankings across all subsets supports the quasi-random model facet assumption (§).

Null calibration. We generate 100 Bernoulli-distributed datasets with item difficulties matched to the empirical distribution but no systematic facet effects. The observed-to-null variance ratio is 1.0008, confirming that Henderson Method I does not introduce systematic bias when applied to binary outcomes. The null simulation also verifies that the item variance share (58.4%) is not an artifact of the Bernoulli $p(1-p)$ ceiling: the null produces item variance of 0.08%, three orders of magnitude below the observed value.

Reduced crossing validation. To verify that the reduced crossing in exp-002 (BF16 only × 2 temperatures) does not introduce confounding, we extracted the corresponding subset from the fully crossed exp-001 design. The generalizability coefficient differs by only 0.0008 ($G = 0.9361$ vs. 0.9369), and all non-precision variance components differ by less than 2 percentage points. The 10pp increase in item variance share (58.4% → 68.5%) is fully explained by the removal of precision-related components (∼4.5% total), which redistributes across remaining sources. This confirms that the exp-002 design preserves the variance structure of the full crossing.

Cross-Model D-Study Projections

Table 5 extends the single-model D-study (Table 4 in main text) to per-benchmark cross-model projections with bootstrap confidence intervals (1,000 item-level resamples, 8 models, 2,304 conditions per item). The efficiency ratio is format-associated: MC benchmarks show 11–13× (mean 12.0× [11.6, 12.5]), while FF benchmarks show 21–22× (mean 21.5× [20.4, 22.8]).

Table 5: Cross-model D-study reproducibility budgets with 95% bootstrap CIs. $n_{G\geq 0.80}$: items needed. Multi = multi-facet (2,304 cond.), Single = single-condition.

	Multi $n_{G\geq 0.80}$		Single $n_{G\geq 0.80}$		Ratio
	Point	95% CI	Point	95% CI	[95% CI]
Multiple-choice					
MMLU	93	[76, 116]	1216	[1004, 1491]	13.1 [12.5, 13.7]
ARC	98	[71, 139]	1078	[793, 1512]	11.0 [10.7, 11.4]
Hella.	118	[93, 151]	1403	[1120, 1790]	11.9 [11.5, 12.3]
Free-form					
GSM8K	128	[103, 162]	2768	[2230, 3531]	21.6 [20.7, 23.0]
MATH	101	[86, 122]	2160	[1853, 2631]	21.4 [20.2, 22.5]

The higher item requirements relative to the single-model analysis (55 items in Table 4) reflect the additional model×item interaction variance that enters σ^2_{rel} when models are crossed as a facet. The format-associated ratio pattern—FF benchmarks showing ∼2× the MC ratio—is consistent with the distributed variance structure observed in §, where FF benchmarks allocate more variance to model and model×prompt components. MATH results use verified per-model inference ensuring complete observations across all 8 models.

Model and Prompt Details

Models. Table 6 lists the eight models with their HuggingFace identifiers.

Table 6: Models used in all experiments.

Short name	HuggingFace identifier	Params
Llama-3.1-8B	`meta-llama/Llama-3.1-8B-Instruct`	8B
Gemma-2-9B	`google/gemma-2-9b-it`	9B
Mistral-7B-v0.3	`mistralai/Mistral-7B-Instruct-v0.3`	7B
Qwen3-8B	`Qwen/Qwen3-8B`	8B
Yi-1.5-9B	`01-ai/Yi-1.5-9B-Chat`	9B
OLMo-2-7B	`allenai/OLMo-2-1124-7B-Instruct`	7B
InternLM2.5-7B	`internlm/internlm2_5-7b-chat`	7B
DeepSeek-7B	`deepseek-ai/deepseek-llm-7b-chat`	7B

Prompt templates. Our six prompt templates follow a surface perturbation design inspired by BrittleBench (Romanou et al. 2026). For multiple-choice benchmarks, templates vary along three dimensions: option layout (lettered vs. numbered), instruction wording (direct vs. contextual), and answer prompt cue. For free-form benchmarks, templates vary chain-of-thought elicitation strategy. All six templates are semantically equivalent—they request the same task with different surface forms—to isolate formatting effects from content effects. ARC-Challenge and HellaSwag use the same six surface patterns as MMLU with domain-appropriate system instructions; MATH mirrors the GSM8K pattern with `\boxed{}` answer formatting.

MMLU templates (representative MC format):

1. `The following are multiple choice questions (with answers) about {subject}.\n\n{question}\nA. ... D. ...\nAnswer:`
2. `Answer the following {subject} questions by selecting the correct option.\nQuestion: {question}\n(A) ... (B) ... (C) ... (D) ...\nCorrect answer:`
3. `Below are {subject} multiple-choice problems. Pick the right letter.\nQ: {question}\n A) ... B) ... C) ... D) ...\nA:`
4. `Answer the following {subject} question by selecting A, B, C, or D.\n{question}\n- A. ... - B. ... - C. ... - D. ...\nAnswer:`
5. `Below is a {subject} exam question with four possible answers. Choose the correct one.\n{question}\nOption A: ... Option B: ... Option C: ... Option D: ...\nCorrect:`
6. `Question ({subject}):\n{question}\nChoices: (A) ... (B) ... (C) ... (D) ...\nYour answer:`

GSM8K templates (representative FF format):

1. `Solve the following math problem step by step.\nQ: {question}\nA: Let's think step by step.`
2. `Answer the following math word problem. Show your work.\nQuestion: {question}\nSolution:`
3. `Solve each problem below. Write your reasoning, then give the final answer after ####.\nProblem: {question}\nReasoning:`
4. `You are a math tutor. Solve the problem and show your steps.\n{question}\nSteps:`
5. `Work through this math question carefully.\nQ: {question}\nWork:`
6. `Math problem:\n{question}\nLet's solve this step by step.`